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Final Report - Group 26

LABORATORIO DI ANALISI DI MISSIONI SPAZIALI

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0 | Introduction

The objective of this project is the design and analysis of an interplanetary mission targeting the orbital insertion of a spacecraft around the asteroid 3908 Nyx (1980 PA) from a geostationary transfer orbit. The primary goal is to minimize the ΔV required for each maneuver, to ensure the maximum level of efficiency in terms of the propellant mass required to carry out the mission.

The study is structured into three main phases, or scenarios, each characterizing a distinct segment of the transfer. Specifically, the mission architecture is structured as follows:

1. FIRST SCENARIO (*Geocentric Transfer*):

This scenario consists of the standard geocentric maneuvers required to transfer from the initial Geostationary Transfer Orbit (GTO) to the designated geocentric parking orbit. In this report, three alternative transfer strategies are analyzed and compared.

2. SECOND SCENARIO (*Heliocentric Transfer*):

This scenario models the direct heliocentric transfer from Earth's orbit to the target orbit of asteroid 3908 Nyx (1980 PA). A numerical optimization is performed to determine the optimal free parameters of this orbit.

3. THIRD SCENARIO (*Hyperbolic escape and arrival*):

This scenario links the geocentric and heliocentric trajectories utilizing the Patched Conics method. This phase is structured into two main tasks:

- The development of the hyperbolic escape orbit from the geocentric parking orbit (Scenario 1) to the planned heliocentric transfer orbit (Scenario 2), analyzed under both coplanar and non-coplanar assumptions.
- The design of the hyperbolic arrival orbit from the heliocentric transfer orbit (Scenario 2) to the final insertion orbit around the target asteroid. A parametric analysis is conducted to evaluate the impact of the arrival geometry on the final orbit insertion maneuver cost.

1 | Scenario 1 - Reaching Parking Orbit

The trajectory of the first scenario is modeled as a sequence of standard impulsive maneuvers with the aim of reaching the state vectors required for the insertion in the final target orbit.

1.1. Orbits Characterization

1.1.1. Initial orbit

The initial Geostationary Transfer Orbit (GTO) is defined by the Keplerian parameters given as input, while the state vector in ECI coordinates is calculated using the `par2car` function.

Keplerian parameters and graphical representation

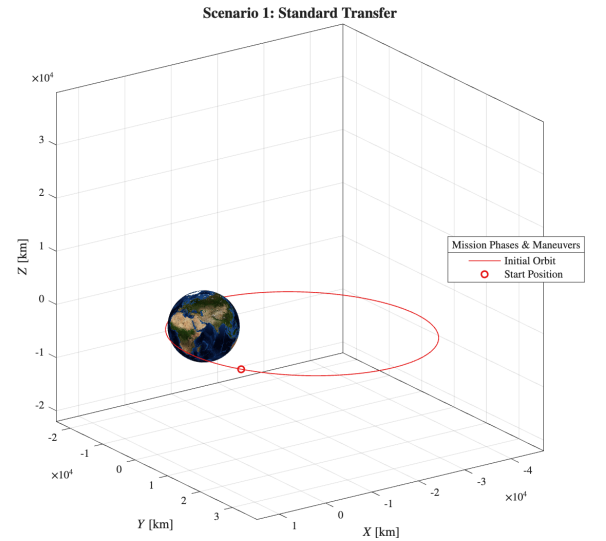
Parameter	Symbol	Value	Unit
Semi-major axis	a_i	24400.00	km
Eccentricity	e_i	0.7283	–
Inclination	i_i	0.1047	rad
RAAN	Ω_i	2.3610	rad
Arg. of pericenter	ω_i	3.1070	rad
True anomaly	θ_i	2.1350	rad

Cartesian vectors at release point

$$\mathbf{r}_i = \begin{pmatrix} 4598.3066 \\ 18115.3697 \\ -1692.5649 \end{pmatrix} \text{ km}, \quad \mathbf{v}_i = \begin{pmatrix} -2.59186 \\ 4.405416 \\ -0.13726 \end{pmatrix} \text{ km/s}$$

1.1.2. Final orbit

The final parking orbit is defined by the state vector in ECI coordinates given as input, while, the Keplerian parameters are calculated using the `car2par` function. Compared to the initial GTO, the final orbit shows differences that require a combined plane change, shape change, and pericenter reorientation manoeuvre sequence.

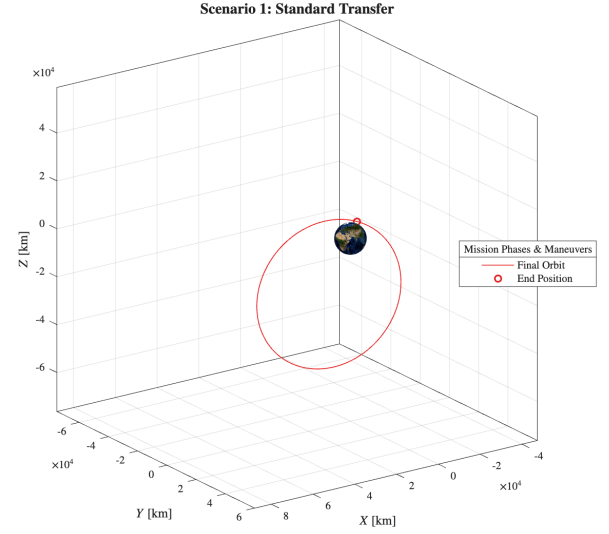


Keplerian parameters and graphical representation

Parameter	Symbol	Value	Unit
Semi-major axis	a_f	42145.32	km
Eccentricity	e_f	0.7659	—
Inclination	i_f	0.4651	rad
RAAN	Ω_f	2.7555	rad
Arg. of pericenter	ω_f	1.1025	rad
True anomaly	θ_f	6.2831	rad

Cartesian vectors at target point

$$\mathbf{r}_f = \begin{pmatrix} -7090.5902 \\ -5612.5573 \\ 3948.9029 \end{pmatrix} \text{ km}, \quad \mathbf{v}_f = \begin{pmatrix} 5.6980 \\ -5.9950 \\ 1.7100 \end{pmatrix} \text{ km/s}$$



1.2. Transfer Trajectory

1.2.1. Problem overview

To complete the transfer at least three maneuvers have to be performed in sequence:

- **Change of orbital plane:** the inclination has to increase from $i_i = 0.1047$ rad to $i_f = 0.4651$ rad, and the RAAN from $\Omega_i = 2.3610$ rad to $\Omega_f = 2.7555$ rad. This is achieved through the `changeOrbitalPlane` function. The maneuver is performed at one of the two intersection nodes between the initial and final planes, which can be selected to minimize the ΔV cost. This maneuver is specifically performed before the argument of pericenter adjustment because it inherently affects this value as a side effect.
- **Change of argument of pericenter:** the argument of pericenter changes from $\omega_i = 3.1070$ rad to $\omega_f = 1.1025$ rad. This is achieved through the `changePericenterArg` function. This apsidal rotation shifts the orientation of the orbit's semi-major axis without changing its size (a) or shape (e). Because it can be performed in two different nodes with the same impulse, the point with the shortest time of flight was always considered.
- **Change of orbit shape:** the semi-major axis increases from $a_i = 24400$ km to $a_f = 42145.32$ km, and the eccentricity from $e_i = 0.7283$ to $e_f = 0.7659$. This can be achieved using either a bi-elliptic transfer (`biellipticTransfer` function) or a bitangent transfer (`bitangentTransfer` function). This second maneuver consists in a first burn, to shape the tangential transfer ellipse, a coast phase, and a last burn to match the final orbit's geometry. However, the transfer can be executed at either of the apsidal points, resulting in four possible strategies: Pericenter-Apocenter (PA), Apocenter-Pericenter (AP), Pericenter-Pericenter (PP), and Apocenter-Apocenter (AA). In the first two strategies, a direct bitangent transfer is used, while in the third, the bi-elliptic principle is explored by using multiple bitangent transfers.

For each of the three strategies, the results in terms of ΔT and ΔV are shown in details in the tables A.1, A.2, A.3. Finally, all the aforementioned maneuvers are modeled under the impulsive approximation, meaning they occur instantaneously.

1.2.2. First strategy - Standard

The standard strategy is the most direct approach to the problem:

1. **Plane change:** performed at the optimal true anomaly $\theta = 210.41\text{deg}$ This position was selected due to its proximity to the apocenter, thereby exploiting the lower orbital velocity to minimize the associated ΔV cost.
2. **Argument of pericenter adjustment:** performed at $\theta = 133.60\text{deg}$ on the initial orbit. This point was selected as the optimal injection point to minimize the time of flight required to reach the target anomaly from the previous orbital position.
3. **Bitangent transfer:** chosen as a PA transfer to minimize the impulse required by leveraging the maximum velocity at the pericenter.

Maneuver	ΔV [km/s]
Plane change	0.8080
Arg. pericenter change	6.2211
Bitangent transfer PA	0.4978
Total	7.5269

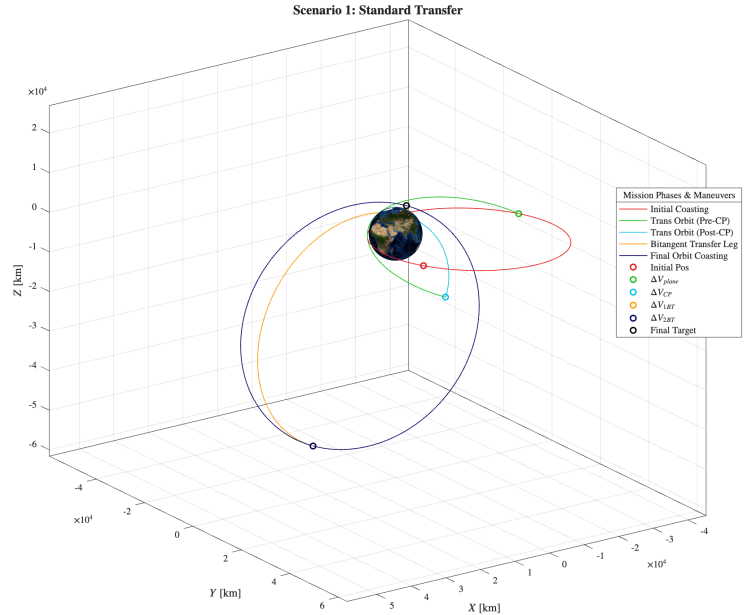


Figure 1.1: Single maneuver costs and graphical representation of the entire first strategy transfer route

The remarkably high cost of the argument of pericenter adjustment ($\Delta V = 6.2211$ km/s) is a result of the substantial difference in the argument of pericenter between the two orbits ($\Delta\omega \approx 2.0$ rad), combined with the fact that this maneuver is executed on the initial GTO, where the local orbital velocity remains relatively high. This second factor also affects the propulsive cost of an orbital plane change which is directly proportional to the magnitude of the velocity vector at the maneuver point. Overall, the total time of flight is:

$$\Delta T = 1.2748 \times 10^5 s$$

1.2.3. Second strategy - ΔT

This strategy rearranges maneuvers to prioritize time-of-flight over propellant consumption:

1. **Bitangent transfer:** chosen as an AP transfer to minimize both the transfer time and the coasting phase on the initial orbit, since the starting position is closer to the apocenter.
2. **The plane change** applied at $\theta = 30.41$ deg, which is the point that can be reached in the shortest time from the current position on the final orbit, determined using the custom `changeOrbitalPlaneDeltaT` function .
3. **Argument pericenter adjustment** at $\theta = 133.60$ deg on the reoriented orbit.

Maneuver	ΔV [km/s]
Bitangent transfer (AP)	0.6452
Plane change	2.9253
Arg. pericenter change	5.3048
Total	8.8753

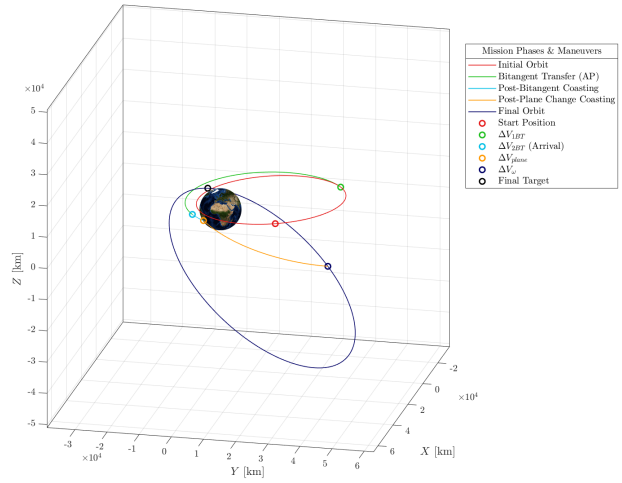


Figure 1.2: Single maneuver costs and graphical representation of the entire second strategy transfer route

The plane change is performed at a true anomaly of $\theta = 30.41$ deg, near the pericenter of the final orbit where the orbital velocity is higher, thereby increasing the cost of this maneuver compared to the previous strategy ($\Delta V_{pc,S1} = 0.8080$ km/s at $\theta_{pc,S1} = 210.41$ deg). Nevertheless, the cost of the argument of pericenter adjustment remains by far the largest contributor, due to similar reasons as in strategy 1. Overall, this is the most expensive strategy in terms of ΔV , but it achieves the shortest time-of-flight:

$$\Delta T = 5.4376 \times 10^4 s$$

1.2.4. Third strategy - ΔV

This strategy exploits the bielliptic transfer principle: unlike a standard direct bitangent transfer it connects the initial and final orbit by introducing an intermediate orbit, where the plane and orientation adjustments are performed at a significantly improved efficiency.

1. **Bitangent transfer (P-A)** to a low-eccentricity intermediate orbit;
2. **Plane change**
3. **Argument pericenter adjustment**
4. **Bitangent transfer (A-P)**

The intermediate orbit parameters ($r_{a,t}$ and e_t) are found through a grid search process.

Maneuver	ΔV [km/s]
Bitangent departure PA	1.5493
Plane change	0.4128
Arg. pericenter	0.0164
Bitangent arrival AP	1.2469
Total	3.2254

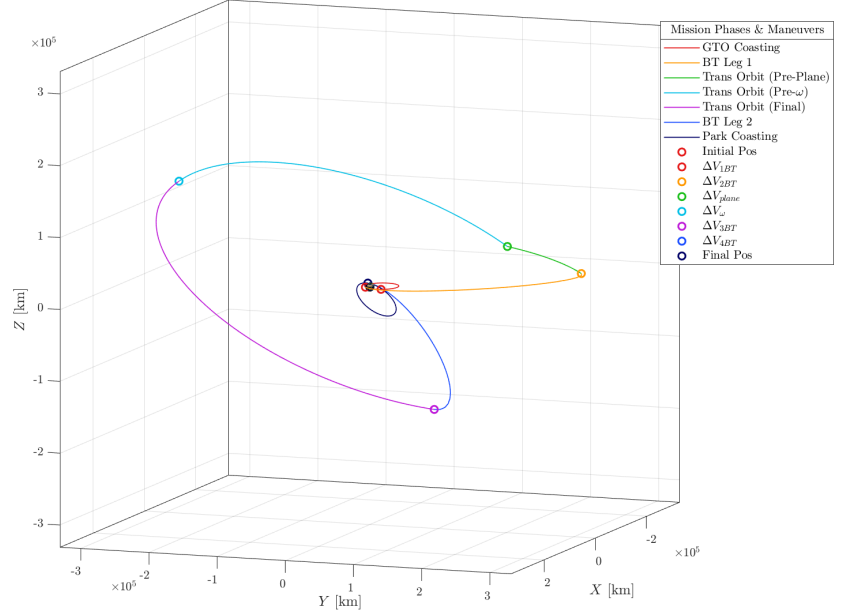


Figure 1.3: Single maneuver costs and graphical representation of the entire first strategy transfer route

The time of flight for this maneuver is: $\Delta T = 2.0619 \times 10^6 s$

1.2.5. Parametric optimization and third strategy analysis

Since infinitely many intermediate orbits could be employed within this bi-elliptic approach, a comprehensive grid search analysis was performed to determine the optimal solution. The optimization space was explored with a grid 500×500 defined by the following bounds: $0.01 \leq e_t \leq 0.95$ and $50000 \leq r_{a,t} \leq 315000$ km. The eccentricity values were constrained to avoid open trajectories in the upper bound, and to prevent a perfectly circular orbit in the lower bound, where the pericenter adjustment argument would no longer be mathematically defined. The intermediate apocenter radius range was chosen to exclude overly low orbits, while ensuring that the maximum altitude remains within the Earth's Sphere of Influence (SOI) to neglect lunar's SOI. The graphics in the next page (1.4 & 1.5) illustrate how the contribution of each individual maneuver scales in relation to these two parameters. The absolute minimum ΔV sits at the domain boundaries: when the intermediate orbit is nearly circular ($e_t = 0.01$) and possesses the largest apocenter radius ($r_{a,t} = 315000$ km).

The physical principles behind this optimal are:

- **Argument pericenter adjustment Optimization:** The argument of pericenter adjustment, which represents the main ΔV penalty in the first two strategies, drops as the eccentricity approaches zero. This behavior is geometrically intuitive: a perfect circle is uniquely defined by its radius and plane, making its spatial orientation within that plane invariant. As a consequence, the impulse required to rotate the line of apsides tends to vanish as shown in fig. 1.5.
- **Plane Change Optimization:** The ΔV required for the plane change maneuver decreases as the apocenter radius increases: by expanding the orbit to the edge of the geocentric SOI, the orbital velocity at the apocenter tends to minimize. Since the impulse required to tilt the orbital plane is proportional to the velocity at the maneuver node, executing the plane change on this

orbit near the apocenter yields substantial advantages as shown in fig. 1.4.

- **Bitangent Transfers Penalty:** The total ΔV for the two sequential bitangent transfers increases with a larger intermediate radius $r_{a,t}$. To reach the optimal $r_{a,t}$, the first burn must inject the spacecraft at the pericenter of a large and elongated ellipse. Conversely, the second burn at the intermediate apocenter decreases because the orbital velocity drops. The subsequent return phase shows the opposite pattern for the same reason, as shown in fig. 1.4.

Therefore, a clear trade-off emerges between the plane change and the combined bitangent transfer cost. The grid search identifies $r_{a,t} = 315000$ km as the optimum because the saving from the plane change outweighs the ΔV penalty for the bitangent transfers.

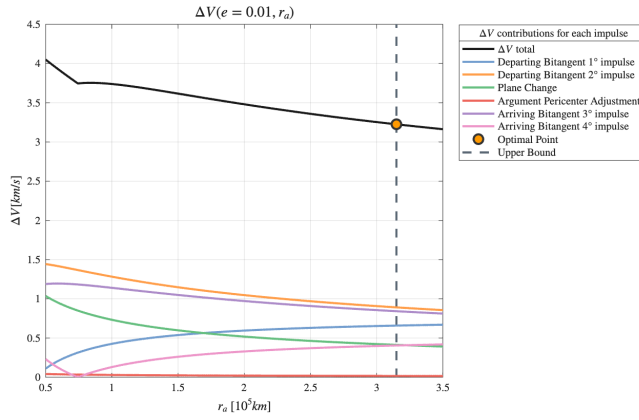


Figure 1.4: ΔV to apocenter radius costs

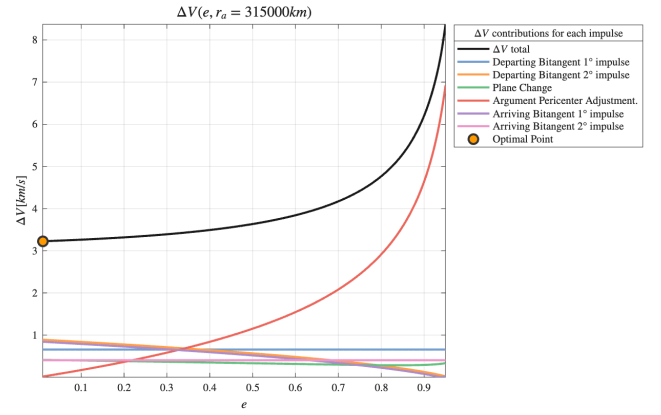


Figure 1.5: ΔV to eccentricity costs

1.2.6. Strategy selection and ΔV contributions graph

Since, the selected goal of this mission is to implement the most efficient maneuvers in terms of ΔV cost, **strategy 3** is selected as the first segment of the mission, despite its significantly longer time-of-flight.

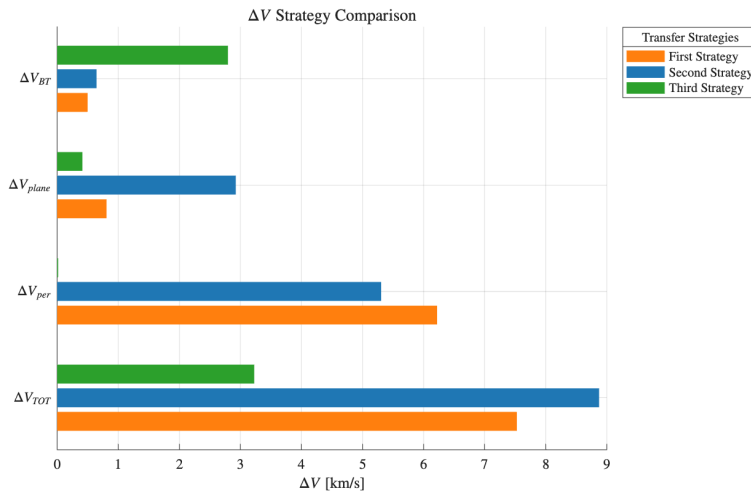


Figure 1.6: Single maneuver costs for every strategy in the first scenario

2 | Scenario 2 - Interplanetary direct transfer to asteroid

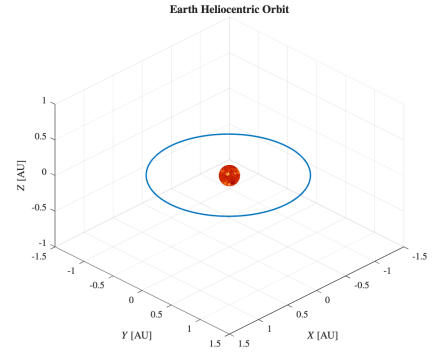
The trajectory of the second scenario is modeled as a direct heliocentric transfer from Earth's orbit to the final orbit of 3908 Nyx.

2.1. Orbits Characterization

2.1.1. Initial orbit: Earth heliocentric orbit

The initial orbit is the Earth's heliocentric orbit defined by the following parameters:

Parameter	Symbol	Value	Unit
Semi-major axis	a_i	1.4946×10^8	km
Eccentricity	e_i	0.016	—
Inclination	i_i	9.192×10^{-5}	rad
RAAN	Ω_i	2.7847	rad
Arg. of pericenter	ω_i	5.2643	rad



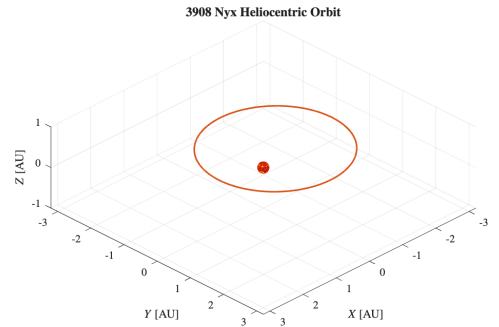
Pericenter & Apocenter: $r_{pi} = a_i(1 - e_i) = 1.4707 \times 10^8$ km $r_{ai} = a_i(1 + e_i) = 1.519 \times 10^8$ km

Orbital speeds: $v_{pi} = \sqrt{\mu_{sun} \left(\frac{2}{r_{pi}} - \frac{1}{a_i} \right)} = 30.2792$ km/s $v_{ai} = \sqrt{\mu_{sun} \left(\frac{2}{r_{ai}} - \frac{1}{a_i} \right)} = 29.3255$ km/s

2.1.2. Final orbit: 3908 Nyx heliocentric orbit

The initial orbit is the Nyx's heliocentric orbit defined by the following parameters:

Parameter	Symbol	Value	Unit
Semi-major axis	a_f	2.884137×10^8	km
Eccentricity	e_f	0.459072	—
Inclination	i_f	0.03822	rad
RAAN	Ω_f	4.5586	rad
Arg. of pericenter	ω_f	2.2106	rad



Pericenter & Apocenter: $r_{pf} = a_f(1 - e_f) = 1.5601 \times 10^8 \text{ km}$ $r_{af} = a_f(1 + e_f) = 4.2082 \times 10^8 \text{ km}$
Orbital speeds: $v_{pf} = \sqrt{\mu_{sun} \left(\frac{2}{r_{pf}} - \frac{1}{a_f} \right)} = 35.2303 \text{ km/s}$ $v_{af} = \sqrt{\mu_{sun} \left(\frac{2}{r_{af}} - \frac{1}{a_f} \right)} = 13.0611 \text{ km/s}$

The gravitational constant used in the heliocentric reference frame is: $\mu_{sun} = 1.3271244 \times 10^{11} \text{ km}^3/\text{s}^2$

2.2. Transfer Trajectory

2.2.1. Problem overview

The objective is to design a direct interplanetary transfer from Earth's initial heliocentric orbit to the final orbit of asteroid 3908 Nyx. The transfer is achieved through a single elliptic orbit intersecting both the initial and final orbits via a two-impulse maneuver: the first impulse inserts the spacecraft into the transfer orbit from the departure point on Earth's orbit, while the second impulse injects it into the final orbit at the arrival point on Nyx's orbit.

To characterize the transfer ellipse, which is not unique, as infinitely many ellipses can connect the two orbits, a range of three parameters was explored: the true anomaly of the departure point on Earth's orbit θ_{1i} , the true anomaly of the arrival point on Nyx's orbit θ_{2f} , and the argument of pericenter of the transfer orbit ω_T , which defines the orientation of the transfer ellipse within its orbital plane.

Once θ_{1i} and θ_{2f} are chosen, the two position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are determined. Their cross product $\mathbf{r}_1 \times \mathbf{r}_2$ uniquely defines the plane of the transfer orbit, fixing its inclination i_T and the right ascension of the ascending node Ω_T . The remaining free parameter ω_T rotates the ellipse within that plane, and together with $\mathbf{r}_1$ and $\mathbf{r}_2$, fully determines the shape parameters a_T and e_T through the polar equation of the conic. Finally, the objective is to find the combination of $(\theta_{1i}, \theta_{2f}, \omega_T)$ that minimizes the total velocity budget:

$$\Delta V_{\text{tot}} = \|\mathbf{v}_{1T} - \mathbf{v}_{1i}\| + \|\mathbf{v}_{2f} - \mathbf{v}_{2T}\|$$

2.2.2. Transfer orbit geometry

Given the three free parameters $(\theta_{1i}, \theta_{2f}, \omega_T)$, for each trial the transfer orbit is characterized through the following steps:

Step 1 – Transfer plane determination. The state vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are defined using the `par2car` function:

$$\begin{aligned} (\mathbf{r}_1, \mathbf{v}_1) &= \text{par2car}(a_i, e_i, i_i, \Omega_i, \omega_i, \theta_i) \\ (\mathbf{r}_2, \mathbf{v}_2) &= \text{par2car}(a_f, e_f, i_f, \Omega_f, \omega_f, \theta_f) \end{aligned}$$

The normal to the transfer orbital plane is then:

$$\hat{\mathbf{h}}_T = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{|\mathbf{r}_1 \times \mathbf{r}_2|}$$

This allows for the definition of the transfer orbit's inclination and right ascension of the ascending node as:

$$i_T = \arccos(\hat{\mathbf{h}}_T \cdot \hat{\mathbf{k}}) \quad \Omega_T = \begin{cases} \arccos(\hat{\mathbf{N}}_T \cdot \hat{\mathbf{i}}) & \text{if } \hat{\mathbf{N}}_T \cdot \hat{\mathbf{j}} \geq 0 \\ 2\pi - \arccos(\hat{\mathbf{N}}_T \cdot \hat{\mathbf{i}}) & \text{if } \hat{\mathbf{N}}_T \cdot \hat{\mathbf{j}} < 0 \end{cases}$$

where $\hat{\mathbf{N}}_T = \frac{\hat{\mathbf{k}} \times \hat{\mathbf{h}}_T}{|\hat{\mathbf{k}} \times \hat{\mathbf{h}}_T|}$ is the unit vector along the line of nodes.

Step 2 – Rotation to the perifocal frame. For a given value of ω_T , the rotation matrix from the heliocentric ecliptic frame to the perifocal frame of the transfer orbit is:

$$\mathbf{T}_{\text{Elio} \rightarrow \text{PF}_{T,n}} = \mathbf{R}_{\omega_T, n} \cdot \mathbf{R}_{i_T} \cdot \mathbf{R}_{\Omega_T}$$

where $\mathbf{R}_{i_T}$ and $\mathbf{R}_{\omega_T}$, $\mathbf{R}_{\Omega_T}$ are the standard rotation matrices about the first and third axes, respectively. The position vectors are then projected into the perifocal frame:

$$\tilde{\mathbf{r}}_1 = \mathbf{T}_{\text{Elio} \rightarrow \text{PF}_{T,n}} \cdot \mathbf{r}_1, \quad \tilde{\mathbf{r}}_2 = \mathbf{T}_{\text{Elio} \rightarrow \text{PF}_{T,n}} \cdot \mathbf{r}_2$$

Step 3 – True anomaly extraction. In the perifocal frame, a point on the ellipse has coordinates $(r \cos \theta, r \sin \theta, 0)$. The true anomalies of the departure and arrival points on the transfer orbit are calculated as:

$$\theta_{1T} = \arctan(\tilde{r}_{1,y}, \tilde{r}_{1,x}), \quad \theta_{2T} = \arctan(\tilde{r}_{2,y}, \tilde{r}_{2,x})$$

Step 4 – Shape parameters from the conic equation. By applying the polar equation of a conic $r = \frac{a(1-e^2)}{1+e \cos \theta}$, it is possible to calculate e_T and a_T :

$$e_T = \frac{r_2 - r_1}{r_1 \cos \theta_{1T} - r_2 \cos \theta_{2T}}$$

$$a_T = \frac{r_1 (1 + e_T \cos \theta_{1T})}{1 - e_T^2}$$

Once these values are determined, only physically valid ones are taken into account:

- closed orbits: $0 \leq e_T < 1$;
- pericenter radius greater than the Sun's radius (with a safety margin): $r_{p,T} > R_{\text{Sun}} + 6.200.000 \text{ km}$.

Step 5 – Velocity computation and ΔV evaluation. Once the orbit is fully defined via $(a_T, e_T, i_T, \Omega_T, \omega_T)$, the velocity vectors at the departure and arrival points on the transfer orbit, $\mathbf{v}_{1T}$ and $\mathbf{v}_{2T}$, can be computed using `par2car`. The total velocity budget is then:

$$\Delta V_{\text{tot}} = \|\mathbf{v}_{1T} - \mathbf{v}_{1i}\| + \|\mathbf{v}_{2f} - \mathbf{v}_{2T}\|$$

where $\mathbf{v}_{1i}$ and $\mathbf{v}_{2f}$ are the heliocentric velocities of Earth and Nyx at their respective encounter points.

2.2.3. Optimization strategy

Using the formulation of the total cost function $\Delta V(\theta_{1i}, \theta_{2f}, \omega_T)$, the primary optimization phase is executed through a parametric grid search analysis. The computational grid combines $100 \times 100 \times 100$ discrete values for each of the three independent parameters, resulting in a total of 10^6 evaluated trajectory combinations.

However, since a grid search only samples a discrete subset of points, the solution obtained by this method alone is considered more as a good preliminary guess rather than the true final optimum. To improve the results a successive continuous optimization is implemented utilizing the Sequential Quadratic Programming (SQP) algorithm within the MATLAB `fmincon` framework. The discrete

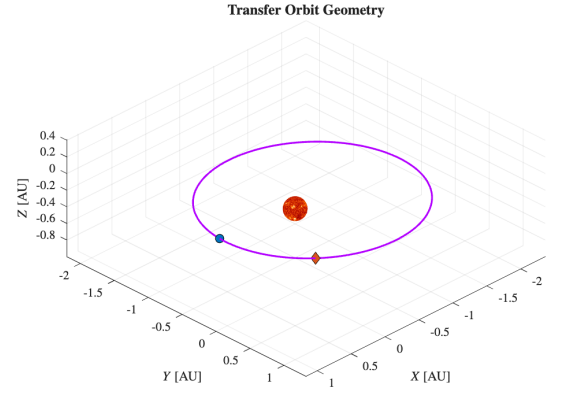
coordinates identified by the grid search are supplied as the initial guess (x_0) for the solver. This specific dual-step methodology ensures high computational robustness: initializing the gradient-based solver already near the globally mapped region mitigates the risk of the SQP algorithm "becoming trapped" in sub-optimal local minimum points, ensuring a more probable convergence toward the absolute minimum.

Table 2.1: Optimization results.

Optimization Phase	θ_{Earth} [deg]	θ_{Nyx} [deg]	ω_T [deg]
Grid Search (Initial Guess)	250.9091	29.0909	7.2727
fmincon (SQP Refinement)	252.1889	28.4243	6.6296

The table that follows, accompanied by the graphical representation, reports the optimal transfer elements:

Parameter	Symbol	Value	Unit
Semi-major axis	a_T	188300132.09	km
Eccentricity	e_T	0.203488	—
Inclination	i_T	1.0365	deg
RAAN	Ω_T	353.4327	deg
Arg. of pericenter	ω_T	6.6295	deg

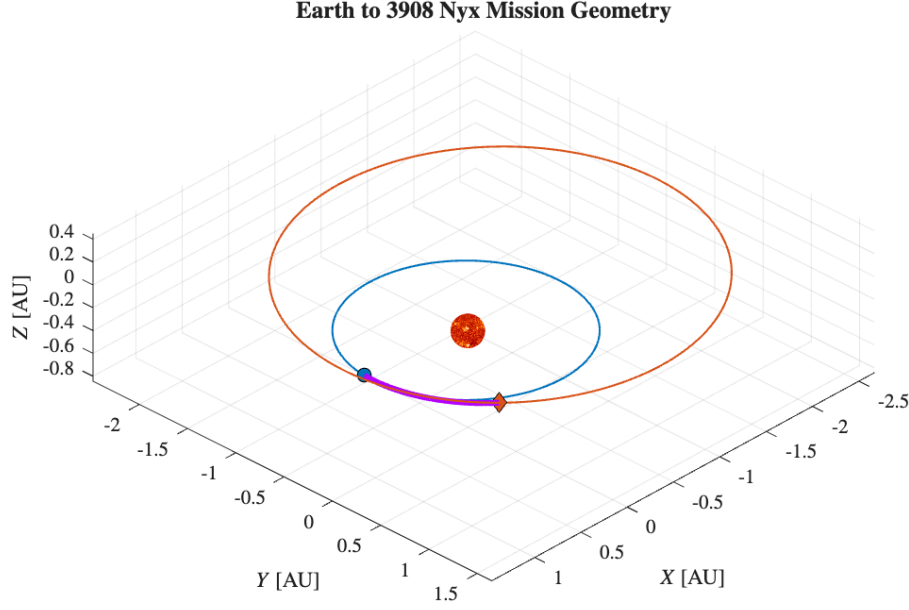


Having defined the geometry of the transfer orbit, the following results characterize the orbit in terms of ΔV performance, in particular we were able to achieve a further gain of 0.003474 km/s thanks to the local optimization.

ΔV	Value
ΔV departure	2.9987 km/s
ΔV arrival	4.0172 km/s
ΔV total (grid search)	7.019371 km/s
ΔV total (fmincon)	7.015898 km/s

The time of flight associated with the optimal transfer orbit, calculated using the TOF_M function , amounts to 61.33 days.

2.2.4. Graphical representation



The cartesian values of the optimal initial and final states for Earth and Nyx are:

$$\mathbf{r}_{\text{Earth}} = \begin{pmatrix} 1.4915 \\ -0.1736 \\ 0 \end{pmatrix} \times 10^8 \text{ km}, \quad \mathbf{v}_{\text{Earth}} = \begin{pmatrix} 2.9769 \\ 29.5101 \\ -0.0026 \end{pmatrix} \text{ km/s}$$

$$\mathbf{r}_{\text{Nyx}} = \begin{pmatrix} 0.8999 \\ 1.3488 \\ 0.0261 \end{pmatrix} \times 10^8 \text{ km}, \quad \mathbf{v}_{\text{Nyx}} = \begin{pmatrix} -25.2398 \\ 23.2036 \\ -1.0897 \end{pmatrix} \text{ km/s}$$

The states at the corresponding boundary points on the transfer orbit are:

$$\mathbf{r}_{T,i} = \begin{pmatrix} 1.4915 \\ -0.1736 \\ 0 \end{pmatrix} \times 10^8 \text{ km}, \quad \mathbf{v}_{T,i} = \begin{pmatrix} 3.1276 \\ 32.4458 \\ 0.5897 \end{pmatrix} \text{ km/s}$$

$$\mathbf{r}_{T,f} = \begin{pmatrix} 0.8999 \\ 1.3488 \\ 0.0261 \end{pmatrix} \times 10^8 \text{ km}, \quad \mathbf{v}_{T,f} = \begin{pmatrix} -22.5634 \\ 20.5617 \\ 0.3229 \end{pmatrix} \text{ km/s}$$

3 | Scenario 3 - Escape and Arrival Phases

In the third scenario, both the departure and arrival hyperbolic trajectories are modeled: the first hyperbola is evaluated under both coplanar and non-coplanar assumptions; while the second one is analyzed with respect to the target insertion altitude around the asteroid.

3.1. Problem overview

In the two previous scenarios, each phase was treated separately, while in an actual mission profile the spacecraft follows a continuous interplanetary trajectory. The exact modeling of the complete path constitutes an N-body problem ($N \geq 3$), involving the simultaneous gravitational attraction of multiple bodies among which the Sun, the Earth, and the asteroid Nyx. Because the N-body problem lacks a closed-form analytical solution for three or more bodies, the equation of motion typically requires a complex numerical approximation. To overcome this challenge, an adopted approximation technique is the Patched Conics method.

3.1.1. The Patched Conics Method

The Patched Conics method is an approximation used to model the spacecraft trajectories within a complex multi-body gravitational environment. Under this framework, the trajectory is divided into multiple segments, each contained within its own **Sphere of Influence (SOI)**. The SOI is defined as the spherical region, centered on a primary celestial body, where its gravitational pull dominates over the secondary body to such an extent that the gravitational effects from other planets can be safely neglected. As a consequence, each phase is considered as a 2-body problem:

1. **Phase 1:** within Earth' SOI following a hyperbolic escape trajectory around the planet.
2. **Phase 2:** within Sun's SOI following the transfer designed in Scenario 2.
3. **Phase 3:** within Nyx's SOI following a hyperbolic arrival trajectory around the asteroid.

To link the different phases, the hyperbolic excess velocity v_∞ , is defined as the difference between the spacecraft velocity on the transfer heliocentric orbit from the second scenario and the velocity of the celestial body around the sun:

$$v_{\infty,1} = \|V_{1T} - V_{Earth}\| = 2.9986 \text{ km/s}$$

$$v_{\infty,2} = \|V_{2T} - V_{Nyx}\| = 4.0172 \text{ km/s}$$

In addition, in the injection point the radius of the hyperbolas must be explicitly set equal to the one from the parking and capturing orbit.

Finally, the SOI radius of a celestial body c is defined as:

$$R_{SOI} = R_{c-\odot} \left(\frac{m_c}{m_\odot} \right)^{2/5}$$

where $R_{c-\odot}$ is the distance between c and the Sun.

In the following table are shown the SOI radius for the Earth and 3908 Nyx:

Body	$R_{c-\odot}$ [km]	m_c [kg]	R_{SOI} [km]
Earth	1.47×10^8	5.972×10^{24}	927979.68
Nyx	2.88×10^8	1.0472×10^{12}	7.9160

3.2. Coplanar Approximation

3.2.1. Escape trajectory: Earth Departure

The escape hyperbola connecting the parking orbit of Scenario 1 to the heliocentric transfer orbit of Scenario 2 can be determined with two input parameters: the hyperbolic excess velocity $v_{\infty,1} = 2.9986$ km/s and the pericenter radius of the parking orbit $r_p = 9867.68$ km.

Under the coplanar approximation performing the maneuver at pericenter of the parking orbit is always the optimal choice as a result of the Oberth effect.

The hyperbola is characterized as follows

$$a_H = -\frac{\mu_{Earth}}{v_{\infty,1}^2}$$

$$e_H = 1 - \frac{r_{pH}}{a_h}$$

$$\Delta_H = -a_H \cdot \sqrt{e_H^2 - 1}$$

$$\delta = 2 \cdot \arcsin \left(\frac{1}{e_H} \right)$$

The velocity at hyperbolic pericenter is obtained from energy conservation:

$$v_{PH} = \sqrt{v_{\infty,1}^2 + \frac{2\mu_{Earth}}{r_p}}$$

The required ΔV is the difference between the hyperbolic pericenter velocity and the orbital velocity at the same point on the parking orbit:

$$\Delta V_1 = |v_{PH} - v_{parking}|$$

Parameter	Symbol	Value	Unit
Hyperbolic excess velocity	v_∞	2.9986	km/s
Semi-major axis	a_H	-44329	km
Eccentricity	e_H	1.2226	—
Pericenter radius	r_p	9867.6797	km
Impact parameter	Δ_H	31180.4568	km
Deflection angle	δ	109.7559	deg
Maneuver ΔV_1		1.0294	km/s

Table 3.1: Earth escape hyperbola parameters

3.2.2. Arrival Trajectory: Nyx Capture

The arrival hyperbolic trajectory connecting the heliocentric transfer orbit to the final circular parking orbit around Nyx is reconstructed analogously to the geocentric departure trajectory: the hyperbolic excess velocity is fixed at $v_{\infty,2} = 4.0173$ km/s, whereas the pericenter radius is treated as a free design variable to be optimized to minimize the ΔV cost.

The optimal value is determined by performing a grid search over a domain bounded by the following operational limits:

- $r_{p,\min} = R_{\text{Nyx}} + 1 = 1.5$ km, representing the physical radius of the asteroid plus an adequate clearance margin to ensure collision avoidance;
- $r_{p,\max} = R_{\text{SOI}} = 7.92$ km, representing the radius of the planetary sphere of influence, imposed to satisfy the requirement of operating within Nyx's gravitational environment.

Due to the extremely low mass of 3908 Nyx, its gravitational parameter is exceptionally small ($\mu_{\text{Nyx}} \approx 6.99 \times 10^{-8}$ km³/s²). The capture ΔV is therefore virtually identical to the hyperbolic excess velocity $v_{\infty,2}$, remaining independent of the chosen pericenter radius.

The pericenter radius is ultimately calculated as:

$$r_p = r_{p,\min} = 1.5 \text{ km}$$

Resulting in a cost of maneuver:

$$\Delta V_2 = |v_{\text{circular}} - v_{PH,2}| = \left| \sqrt{\frac{\mu_{\text{Nyx}}}{r_p}} - \sqrt{v_{\infty,2}^2 + \frac{2\mu_{\text{Nyx}}}{r_p}} \right| = 4.0170 \text{ km/s}$$

Parameter	Symbol	Value	Unit
Hyperbolic excess velocity	v_∞	4.0170	km/s
Semi-major axis	a_H	-4.330787×10^{-9}	km
Eccentricity	e_H	1.3854×10^8	–
Pericenter radius	r_p	1.5	km
Impact parameter	Δ_H	1.5	km
Deflection angle	δ	8.2712×10^{-7}	deg
Maneuver ΔV_2		4.0170	km/s

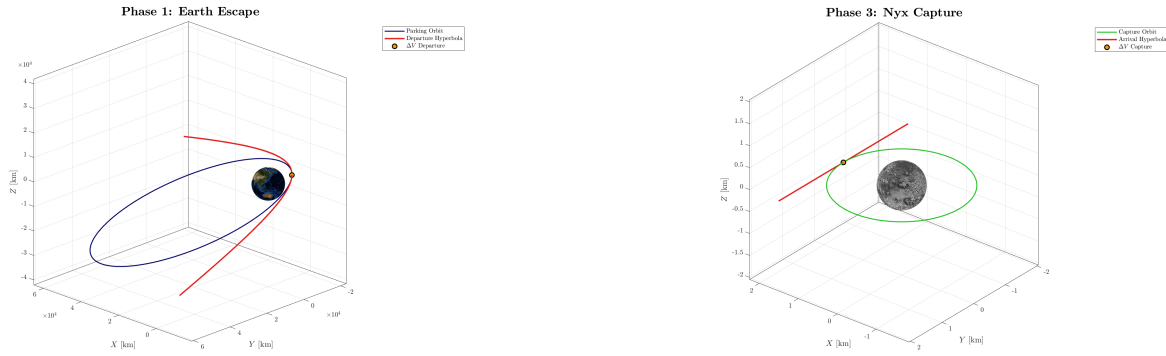
Table 3.2: Nyx arrival hyperbola parameters

3.2.3. Total ΔV budget

The Earth departure phase accounts for 1.0294 km/s to the mission budget, whereas the Nyx arrival and capture maneuver requires 4.0170 km/s. Therefore, the total impulse requirement obtained from the patched-conics approximation, under the coplanar assumption, is:

$$\Delta V_{\text{tot}} = \Delta V_{\text{Earth escape}} + \Delta V_{\text{Nyx capture}} = 5.0463 \text{ km/s}$$

3.2.4. Graphical representation



3.3. Non-coplanar case: Plane Change at Departure

To determine the non-coplanar departure hyperbola, the asymptote of the escape trajectory must be coincident with the direction of the hyperbolic excess velocity vector ($\vec{v}_\infty$). This vector was then converted from the Heliocentric Ecliptic (ECLIP) frame to the Geocentric Equatorial Inertial (ECI) coordinate system.

To find the geometric parameters of the departure hyperbola, the following system of equations must be solved:

$$\begin{cases} r_h = \frac{a_h(1-e_h^2)}{1+e_h \cos(\theta_h)} \\ \cos(\theta_\infty) = -\frac{1}{e_h} \\ \theta_\infty = \alpha + \theta_h \end{cases}$$

Where r_h is the free parameter, the hyperbolic semi-major axis is given by $a_h = -\frac{\mu}{\|\vec{v}_\infty\|^2}$ due to the conservation of energy, and the angle α is defined by the geometry between the position vector at the

injection burn and the escape asymptote:

$$\alpha = \arccos(\hat{r}_h \cdot \hat{v}_\infty)$$

This system was solved numerically utilizing the `fsolve` function implemented in MATLAB, after performing analytical simplifications to improve convergence. Under non-coplanar conditions, the pericenter of the initial orbit is no longer guaranteed to be the optimal departure location: virtually any point along the starting orbit can represent a valid injection node, provided that it satisfies the following physical and operational constraints:

$$\begin{cases} \frac{\pi}{2} < \theta_\infty < \pi \\ r_{p,h} > R_\oplus \end{cases}$$

where the first condition ensures a valid hyperbolic trajectory, and the second guarantees that the resulting hyperbolic pericenter radius ($r_{p,h}$) is larger than Earth's physical radius ($R_\oplus$).

Once the planar parameters (a_h , e_h , and θ_h) are determined, the three plane orientation of the departing hyperbola must be computed.

First, the angular momentum vector ($\hat{h}$), perpendicular to the hyperbolic trajectory plane, is determined via the cross product between the injection position vector ($\hat{r}_h$) and the escape asymptote direction ($\hat{r}_\infty$):

$$\hat{h} = \frac{\hat{r}_h \times \hat{r}_\infty}{\|\hat{r}_h \times \hat{r}_\infty\|} \quad \vec{h}_h = \sqrt{\mu a_h (1 - e_h^2)} \hat{h}$$

Geometrically, the cross product alone introduces an ambiguity about the direction of movement in the orbital plane. To solve this problem, an iterative verification loop is implemented to check the direction of motion along the hyperbola by confronting the direction of the velocity at injection point with the one of the excess velocity:

$$\vec{v}_{hyp} \cdot \vec{v}_\infty$$

If this scalar product is negative the computed orbit would wrap around the Earth in the wrong direction and the direction angular momentum must be inverted:

Then the node vector $\vec{N}_h$ is computed as:

$$\vec{N}_h = \frac{\hat{k} \times \vec{h}_h}{\|\hat{k} \times \vec{h}_h\|}$$

where $\hat{k} = [0, 0, 1]^T$.

The inclination is uniquely determined by:

$$i_h = \arccos\left(\frac{\vec{h}_h \cdot \hat{k}}{\|\vec{h}_h\|}\right)$$

The Right Ascension of the Ascending Node (Ω_h) is calculated as follows:

$$\Omega_h = \begin{cases} \arccos(N_{x,h}) & \text{if } N_{y,h} \geq 0 \\ 2\pi - \arccos(N_{x,h}) & \text{if } N_{y,h} < 0 \end{cases}$$

Defining the angle β from the line of nodes to the position vector::

$$\beta = \arctan \left((\vec{N}_h \times \hat{r}_h) \cdot \hat{h}, \vec{N}_h \cdot \hat{r}_h \right)$$

Since the injection point is located at a true anomaly of θ_h on the hyperbola, the argument of pericenter is obtained as:

$$\omega_h = \beta - \theta_h$$

In order to determine the absolute optimal injection point, a parametric grid search analysis was performed. The search space evaluated the true anomaly ($\theta_{h,in}$) of the injection node on the initial orbit, from 0 deg to 360 deg with a step size sensitivity of 1 deg. Finally, for each point, the non-coplanar departure hyperbola was reconstructed and the corresponding total insertion ΔV was computed. The optimal departure point on the initial orbit is $\theta_{h,in} = 165$ deg and the optimal hyperbola parameters are shown in the following table:

Parameter	Symbol	Value	Unit
Hyperbolic excess velocity	v_∞	2.9986	km/s
Eccentricity	e_H	2.3277	—
Inclination	i_H	45.43	deg
RAAN	Ω_H	43.27	deg
Pericenter argument	ω_H	297.80	deg
Periapsis radius	r_p	58849.63	km
Impact parameter	Δ_H	93168.82	km
Deflection angle	δ	50.89	deg
Maneuver ΔV_2		3.8304	km/s

Table 3.3: Earth escape non coplanar hyperbola parameters

3.3.1. Graphical representation

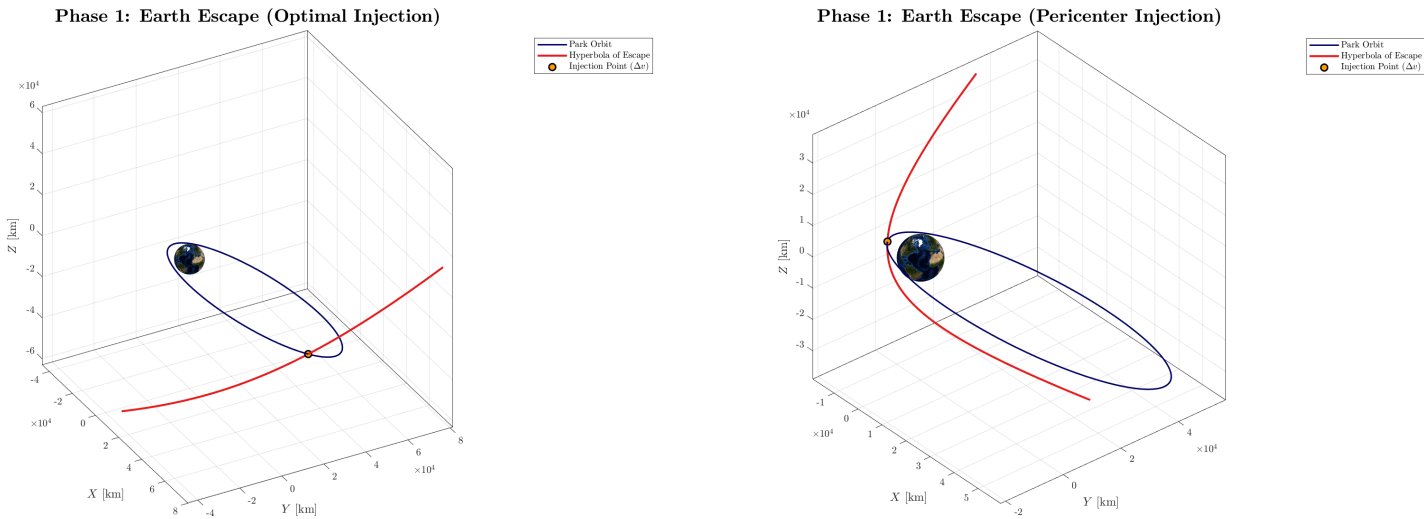


Figure 3.1: Hyperbolic escape orbits for an injection at the optimal point and at the pericenter of the parking orbit

4 | Conclusions

In this report, the feasibility and design of an interplanetary transfer from Earth to the asteroid 3908 Nyx were analyzed. Throughout the study, the primary objective was to identify the most efficient transfer to minimize the total ΔV cost .

This goal was achieved successfully through an optimization process. The numerical results demonstrate significant propulsive savings: about 57% in the first scenario (through the bi-elliptic Strategy 3) as shown in 4.1 and 74% in the third scenario (by selecting the absolute optimal injection node in the non-coplanar hyperbola).

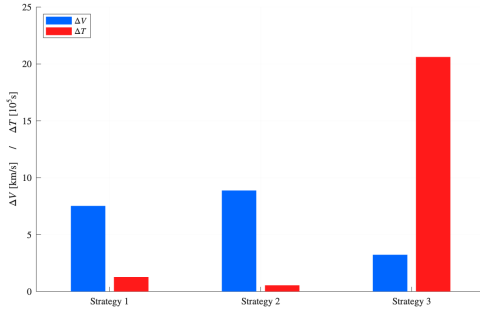


Figure 4.1: ΔV and ΔT Comparison for Scenario 1 strategies

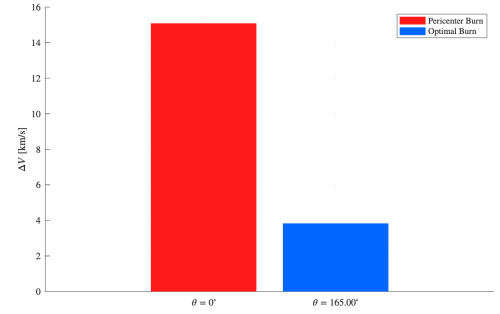


Figure 4.2: ΔV Comparison for injection point in Scenario 3

The overall propulsive cost required for the entire interplanetary transfer corresponds to $\Delta V = 11.0729 \text{ km/s}$, while the overall time of flight is 85 days 12 hours 11 minutes 30 seconds.

Phase	$\Delta V[\text{km/s}]$	$\Delta T[10^4 \text{s}]$
Scenario 1	3.2254	206.188
Scenario 2	0**	529.918
Scenario 3 (dep.)	3.8304	2.683
Scenario 3 (arr.)	4.0171	-
Total	11.0729	738.789

Note:** The ΔV contribution for Scenario 2 is implicitly accounted inside the hyperbolic excess velocities ($\vec{v}_\infty$) of both the departure and arrival trajectories and it is not added as a separate component in the total calculation.

The resulting transfer trajectory must be interpreted as a preliminary theoretical study, rather than an operational mission profile, as it relies on several approximations:

- **Impulsive Maneuvers:** All propulsive burns were modeled as instantaneous velocity increments (ΔV).
- **Two-Body Problem Framework:** The complex multi-body gravitational environment was segmented according to the Patched Conics Approximation. This framework reduces each mission phase to a sequence of isolated two-body problems governed by a single dominant attractor at a time.
- **Absence of orbit corrective maneuvers:** Trajectory corrections were entirely neglected. In a real-world scenario, statistical mid-course correction burns are mandatory to counteract errors and perturbations, which are particularly critical for maintaining stability e.g. along low-eccentricity trajectories (such the one selected for Scenario 1).
- **Simplified asteroid modeling:** The target asteroid 3908 Nyx was modeled as a perfect sphere with a uniform mass distribution. In reality, irregular geometries and mass asymmetries generate non-spherical gravitational harmonics that drastically alter the orbital dynamics in the proximity of the asteroid.
- **Idealized Space Environment:** The geocentric phase were assumed to be free of space debris, micro-meteoroids, and secondary spacecraft, thereby neglecting operational collision-avoidance constraints.

Future developments of this work could involve multi-objective optimization algorithms. These frameworks may focus on the time-of-flight (ΔT) as a primary performance metric, explore constrained optimization problems (e.g. minimizing ΔV subject to a maximum ΔT), or minimize individual peak impulses rather than the integrated total cost. Such variations would broaden the selection envelope for alternative propulsion technologies, different spacecraft architectures or mission requirements.

Bibliography

A | Appendix A

Tables of the orbital parameters of all the designed orbits.

A.0.1. Scenario 1

First strategy

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	24400	0.7283	0.1047	2.3610	3.1070	2.1350	-
26957.0	24400	0.7283	0.1047	2.3610	3.1070	3.6723	0.8080
-	24400	0.7283	0.4651	2.7555	2.7220	3.6723	-
39196.1	24400	0.7283	0.4651	2.7555	2.7220	2.3318	6.2211
-	24400	0.7283	0.4651	2.7555	1.1025	3.9513	-
43826.5	24400	0.7283	0.4651	2.7555	1.1025	0	0.4978
84422.7	42145.3	0.7659	0.4651	2.7555	1.1025	3.1416	-
127475	42145.3	0.7659	0.4651	2.7555	1.1025	6.2831	-

Table A.1: Scenario 1 - Transfer Strategy 1

Main attractor: Earth

Reference frame: Geocentric Inertial

Second strategy

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	24400	0.7283	0.1047	2.3610	3.1070	2.1350	-
15600.1	24400	0.7283	0.1047	2.3610	3.1070	3.1416	0.6452
36484.3	42145.3	0.7659	0.1047	2.3610	3.1070	0	-
37130.9	42145.3	0.7659	0.1047	2.3610	3.1070	0.5308	2.9253
-	42145.3	0.7659	0.4651	2.7555	2.7220	0.5308	-
45430.4	42145.3	0.7659	0.4651	2.7555	2.7220	2.3318	5.3048
-	42145.3	0.7659	0.4651	2.7555	1.1025	3.9513	-
54376.3	42145.3	0.7659	0.4651	2.7555	1.1025	6.2831	-

Table A.2: Scenario 1 - Transfer Strategy 2

Main attractor: Earth

Reference frame: Geocentric Inertial

Third strategy

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	24400	0.7283	0.1047	2.3610	3.1070	2.1350	-
34565.6	24400	0.7283	0.1047	2.3610	3.1070	0	1.5493
355465.7	315000	0.01	0.1047	2.3610	3.1070	3.1416	-
504714.6	315000	0.01	0.1047	2.3610	3.1070	3.6724	0.4128
-	315000	0.01	0.4651	2.7555	2.722	3.6724	-
1002741	315000	0.01	0.4651	2.7555	2.722	5.4734	0.0164
-	315000	0.01	0.4651	2.7555	1.1025	0.8098	-
1650016	315000	0.01	0.4651	2.7555	1.1025	3.1416	1.2469
1975774	42145.3	0.7659	0.4651	2.7555	1.1025	0	-
2061881	42145.3	0.7659	0.4651	2.7555	1.1025	6.2831	-

Table A.3: Scenario 1 - Transfer Strategy 3

Main attractor: Earth

Reference frame: Geocentric Inertial

A.0.2. Scenario 2

t [s]	a [10^8 km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	1.4960	0.016	$9.1920e-05$	2.7847	5.2643	4.4015	-
-	1.4960	0.016	$9.1920e-05$	2.7847	5.2643	4.4015	2.9986
-	1.8830	0.2035	0.018	6.1686	0.1157	6.1663	-
5299175	1.8830	0.2035	0.018	6.1686	0.1157	0.9814	4.0173
-	2.8841	0.4591	0.0382	4.5586	2.2106	0.4961	-
5299175	2.8841	0.4591	0.0382	4.5586	2.2106	0.4961	-

Table A.4: Scenario 2 - Interplanetary Transfer (Earth to Nyx)

Main attractor: The Sun

Reference frame: Heliocentric Inertial

A.0.3. Scenario 3

Coplanar Scenario

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	42145.3	0.7659	0.4651	2.7555	1.1025	0	-
-	42145.3	0.7659	0.4651	2.7555	1.1025	0	1.0295
-	-44326.1	1.2226	0.4651	2.7555	1.1025	0	-
-	-44326.1	1.2226	0.4651	2.7555	1.1025	0	-

Table A.5: Scenario 3 - From Earth's parking orbit to hyperbolic escape trajectory

Main attractor: Earth

Reference frame: Geocentric Inertial

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	$-4.3307e-9$	$3.4636e+08$	0.0382	4.5586	2.2106	0	-
-	$-4.3307e-9$	$3.4636e+08$	0.0382	4.5586	2.2106	0	4.0171
-	1.5	0	0.0382	4.5586	2.2106	0	-
-	1.5	0	0.0382	4.5586	2.2106	0	-

Table A.6: Scenario 3 - From Nyx's hyperbolic arrival trajectory to a circular parking orbit
Main attractor: Nyx 3908
Reference frame: Inertial

Non coplanar Scenario

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	42145.3	0.7659	0.4651	2.7555	1.1025	0	-
-	42145.3	0.7659	0.4651	2.7555	1.1025	0	15.0867
-	-44326.1	1.1928	2.1927	4.1296	6.0811	0.7168	-
-	-44326.1	1.1928	2.1927	4.1296	6.0811	0.7168	-

Table A.7: Scenario 3 - From Earth's parking orbit to hyperbolic escape trajectory with injection
at the parking orbit's pericenter
Main attractor: Earth
Reference frame: Geocentric Inertial

t [s]	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	θ [rad]	ΔV [km/s]
0	42145.3	0.7659	0.4651	2.7555	1.1025	0	-
26832.3	42145.3	0.7659	0.4651	2.7555	1.1025	2.8798	3.8304
-	-44326.1	2.3277	0.7929	0.7552	5.1976	0.5973	-
-	-44326.1	2.3277	0.7929	0.7552	5.1976	0.5973	-

Table A.8: Scenario 3 - From Earth's parking orbit to hyperbolic escape trajectory with injection
at the optimal point of the parking orbit
Main attractor: Earth
Reference frame: Geocentric Inertial